## A8.2 Multiple region signaling

This section describes the optional additional signaling in the AXI4 protocol to support multiple region interfaces.

### A8.2.1 Additional interface signals

Optionally, the AXI4 interface signal set can be extended to support two 4-bit region identifiers:

**AWREGION** A region identifier, sent on the write address channel for each write transaction.

**ARREGION** A region identifier, sent on the read address channel for each read transaction.

In this specification, **AxREGION** indicates **AWREGION** or **ARREGION**.

The 4-bit region identifier can be used to uniquely identify up to sixteen different regions. The region identifier can provide a decode of higher order address bits. The region identifier must remain constant within any 4Kbyte address space.

The use of region identifiers means a single physical interface on a slave can provide multiple logical interfaces, each with a different location in the system address map. The use of the region identifier means that the slave does not have to support the address decode between the different logical interfaces.

This protocol expects an interconnect to produce **AxREGION** signals when performing the address decode function for a single slave that has multiple logical interfaces. If a slave only has a single physical interface in the system address map, the interconnect must use the default **AxREGION** values. See Chapter A10 *Default Signaling and Interoperability*.

There are a number of usage models for the region identifier, including but not limited to the following:

- A peripheral can have its main data path and control registers at different locations in the address map, and be accessed through a single interface without the need for the slave to perform an address decode.
- A slave can exhibit different behaviors in different memory regions. For example, a slave might provide read and write access in one region, but read only access in another region.

A slave must ensure the correct protocol signaling and the correct ordering of transactions are maintained. A slave must ensure that it provides the responses to two transactions to different regions with the same AXI ID in the correct order.

A slave must also ensure the correct protocol signaling for any values of **AxREGION**. If a slave implements less than sixteen regions, then the slave must ensure the correct protocol signaling on any attempted access to an unsupported region. How this is achieved is IMPLEMENTATION DEFINED. For example, the slave might ensure this by:

- providing an error response for any transaction that accesses an unsupported region
- aliasing supported regions across all unsupported regions, to ensure a protocol-compliant response is given for all accesses.

The **AxREGION** signals only provide an address decode of the existing address space that can be used by slaves to remove the need for an address decode function. The signals do not create new independent address spaces. **AxREGION** must only be present on an interface that is downstream of an address decode function.


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